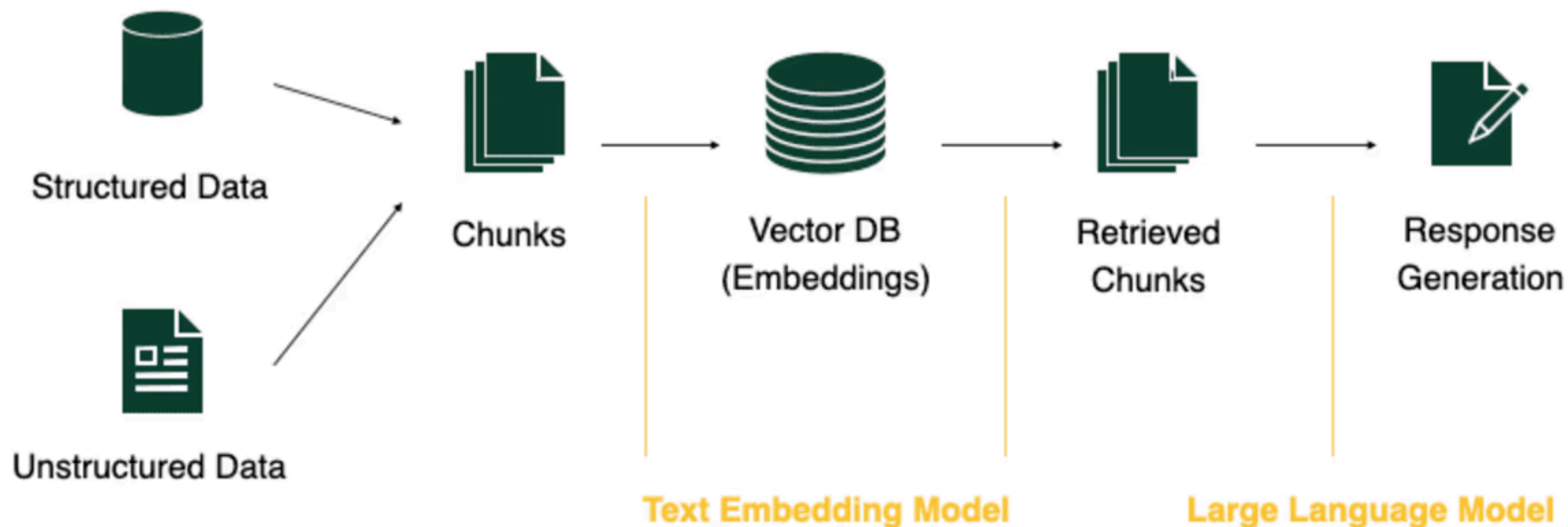


GraphRAG



GraphRAG (Graph-enhanced Retrieval-Augmented Generation) is an advanced AI method that combines large language models (LLMs) with knowledge graphs to retrieve and process information. Unlike traditional RAG—which relies only on text similarity to find separate data chunks—GraphRAG organizes information into a network of nodes (entities) and edges (relationships). This structure lets the AI follow connections across multiple data points to answer complex, multi-step questions.

Simple RAG

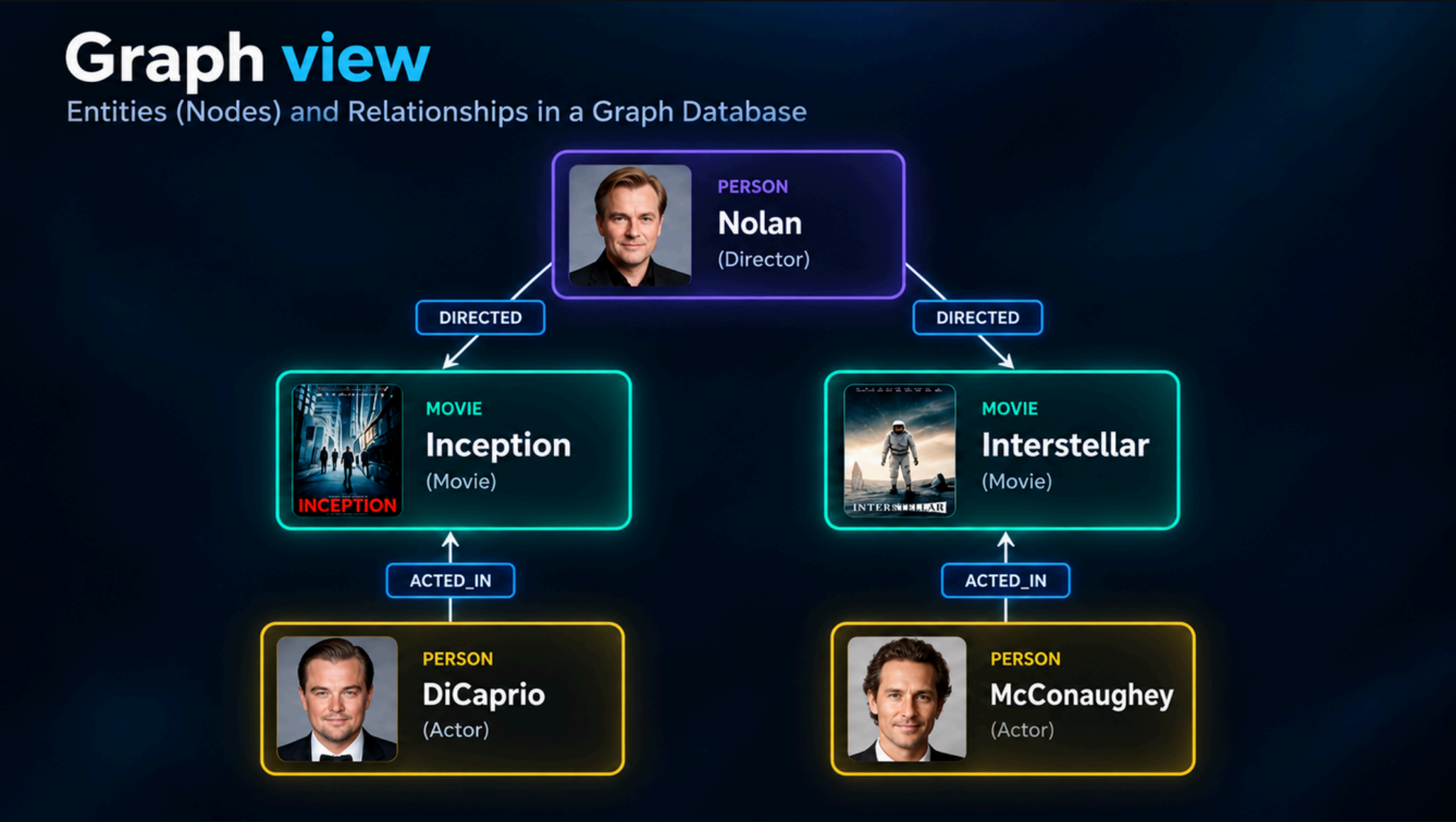


Why a graph?

Because many real questions are about connections, not isolated rows.

Traditional table view

Movie	Director	Actor
Inception	Christopher Nolan	Leonardo DiCaprio
Interstellar	Christopher Nolan	Matthew McConaughey





Neo4j is a native graph database designed to store, manage, and query highly connected data using nodes, relationships, and properties. Instead of organizing information into traditional rows and columns like relational databases, Neo4j saves data the way you sketch it on a whiteboard, making it exceptionally fast for handling complex relationships.

Neo4j: The 5 things you need to know

Almost everything in our practical demo comes back to these five ideas.



1. Node

A real thing.
Example: a Movie or Person.



2. Label

The node type.
Example: :Movie,
:Person.



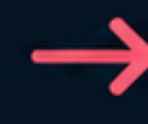
3. Property

Details about a node.
Example: title, year,
name.



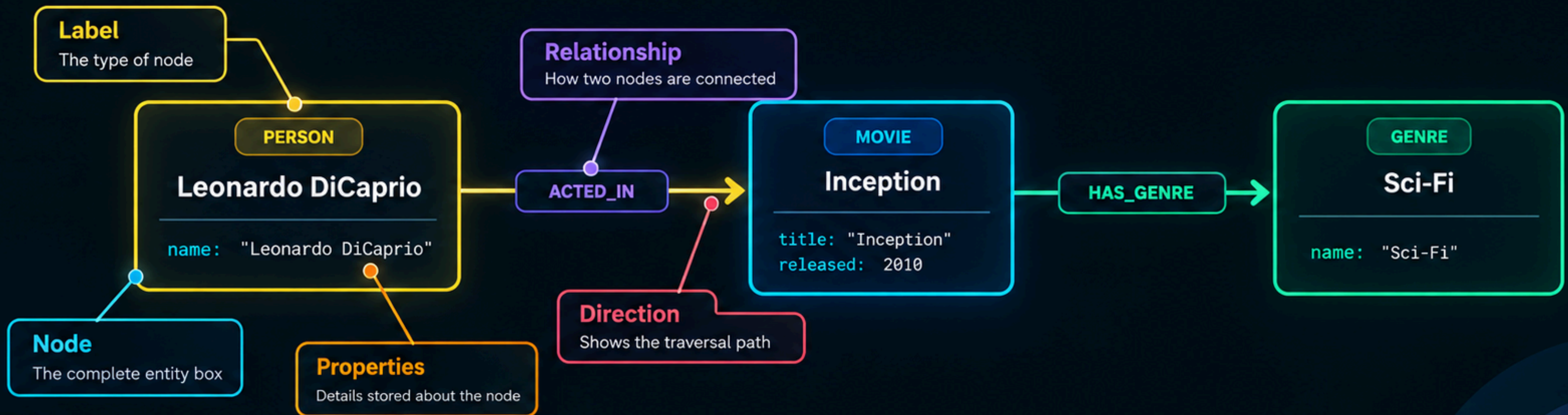
4. Relationship

How nodes are
connected.
Example: ACTED_IN.



5. Direction

Who connects to whom.
Person → Movie.



Cypher: How we talk to the graph

Cypher is Neo4j's query language. Think in patterns instead of joins.

CREATE

Create a node or relationship.

```
CREATE (:Movie {title:"Inception"})
```

MATCH

Find a graph pattern.

```
MATCH (m:Movie) RETURN m
```

MATCH + Relationship

Traverse connections.

```
MATCH (p)-[:ACTED_IN]->(m) RETURN p,m
```

Read this pattern like a sentence:

(Person) —[:ACTED_IN]→ (Movie)

“Find a Person who ACTED_IN a Movie.”

Example:

We will use only three short documents so the whole pipeline stays easy to follow.

Document 1

“Inception is a 2010 science-fiction movie directed by Christopher Nolan. Leonardo DiCaprio acted in the movie.”

Document 2

“Interstellar is a 2014 science-fiction movie directed by Christopher Nolan. Matthew McConaughey acted in the movie.”

Document 3

“The Dark Knight is a 2008 action movie directed by Christopher Nolan. Christian Bale acted in the movie.”

What information is hidden inside these sentences?

Entities

Movies • People • Genres

Relationships

DIRECTED • ACTED_IN • HAS_GENRE

Properties

name • title • released

Step 1 — Break documents into chunks

RAG does not usually send an entire knowledge base to the LLM at once.



Step 2 — Extract entities and relationships

An LLM can transform each chunk into structured graph facts.

Chunk

“Inception is a 2010 science-fiction movie directed by Christopher Nolan. Leonardo DiCaprio acted in the movie.”

LLM

Extracted graph facts

Nodes

- Movie: Inception (released: 2010)
- Person: Christopher Nolan
- Person: Leonardo DiCaprio
- Genre: Sci-Fi

Relationships

- Christopher Nolan —DIRECTED→ Inception
- Leonardo DiCaprio —ACTED_IN→ Inception
- Inception —HAS_GENRE→ Sci-Fi

Step 3 — Build the graph schema

The schema is simply the kinds of nodes and connections we allow.

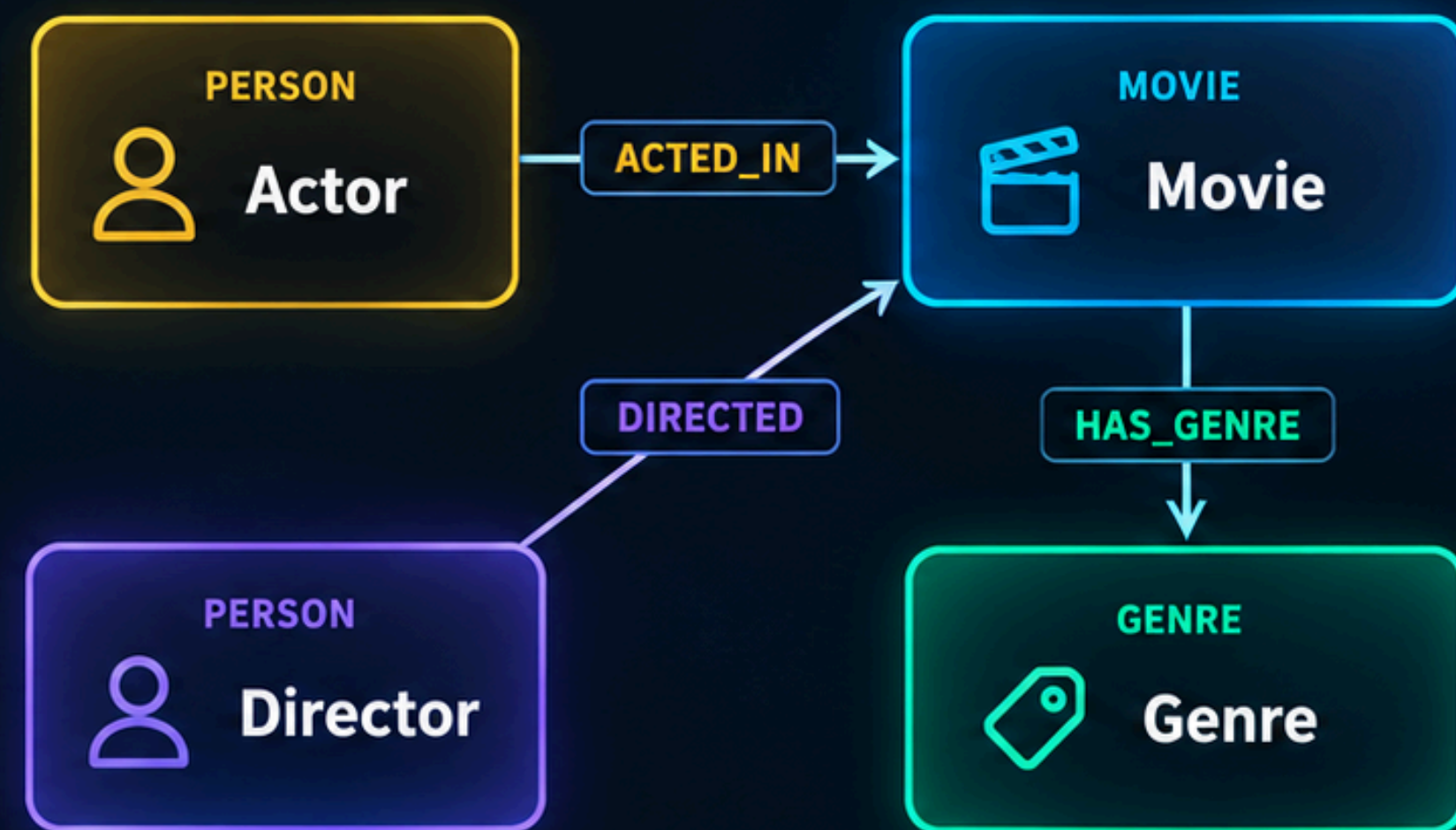
Node labels



Relationship types



Simple pattern

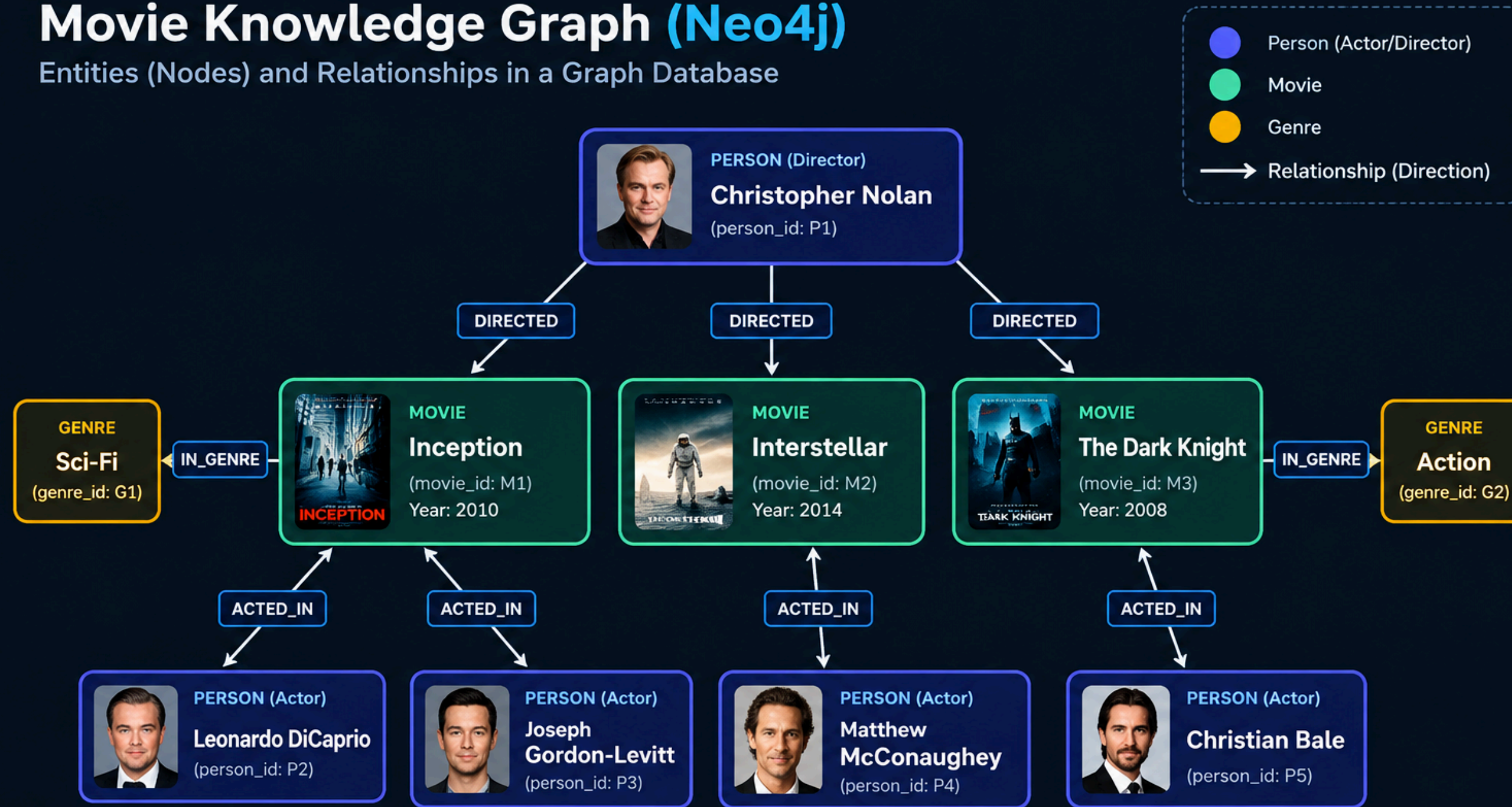


For a beginner demo, keep the schema small. You can always add more node and relationship types later.

Step 4 — Store the graph in Neo4j

Movie Knowledge Graph (Neo4j)

Entities (Nodes) and Relationships in a Graph Database



Example Queries

1. Who directed Inception?
→ Christopher Nolan

2. Which movies did Leonardo DiCaprio act in?
→ Inception

3. What is the genre of Interstellar?
→ Sci-Fi

Normal RAG vs GraphRAG

Both retrieve context for an LLM. GraphRAG adds explicit relationships.

Normal RAG

1 Question
“Who directed Inception?”

2 Vector search
Find similar text chunks.

3 LLM answer
Answer from retrieved text.

GraphRAG

1 Question
Ask about connected facts.

2 Graph retrieval
Traverse nodes + relationships.

3 LLM answer
Answer with connected context.

Questions
+ Graphs
= Deeper Answers

How GraphRAG Retrieval Works

From user question to Cypher query to graph retrieval to final LLM answer

Real
Connections
Real Insights

1 User Question

A user asks a question in natural language.



"Who acted in movies directed by Christopher Nolan?"

2 LLM understands the question

The LLM extracts key entities and understands the intent.



Identified from the question:



Christopher Nolan



acted in



movies

3 LLM reads Neo4j schema

The LLM looks at the graph schema to understand how the data is structured.

Graph Schema (Neo4j)

Nodes

:Person

:Movie

:Genre

Relationships

:DIRECTED

:ACTED_IN

:HAS_GENRE

The schema helps the LLM know which nodes, relationships and properties to use.

4 Generate Cypher Query

The LLM generates a Cypher query based on the question and schema.

Cypher Query

```
MATCH (d:Person)-[:DIRECTED]->(m:Movie)
MATCH (a:Person)-[:ACTED_IN]->(m)
WHERE d.name = "Christopher Nolan"
RETURN m.title, a.name
```

5 Neo4j Graph Retrieval

The query is executed on Neo4j to find matching nodes and relationships in the graph.



Christopher Nolan
(Director)

:Person

DIRECTED



Inception
(Movie)

:Movie

ACTED_IN



Leonardo DiCaprio
(Actor)

:Person

DIRECTED



Interstellar
(Movie)

:Movie

ACTED_IN



Matthew McConaughey
(Actor)

:Person

6 Retrieved Graph Context

Structured results from the graph that will be given to the LLM as context.



Retrieved Results



Inception

→ Leonardo DiCaprio



Interstellar

→ Matthew McConaughey

7 LLM Generates Final Answer

The LLM uses the retrieved graph context to generate a natural language answer.



Final Answer

Christopher Nolan directed Inception and Interstellar. Leonardo DiCaprio acted in Inception, and Matthew McConaughey acted in Interstellar.



Retriever in GraphRAG = Cypher generation + graph traversal + retrieved connected facts

From Data
to Deeper Understanding

The complete GraphRAG pipeline

Two phases: build the knowledge graph once, then query it many times.

INGESTION / BUILD



QUERY / ANSWER

